

Bepaal die volgende integrale / Find the following integrals

1.  $\int (1-3x)^2(3+x)^2 dx$
2.  $\int x^3 \sqrt{1-16x^4} dx$
3.  $\int e^{\sin(4t)} \cos(4t) dt$
4.  $\int \sin(2^{3x}) 2^{3x} dx$
5.  $\int \cos \pi dx$
6.  $\int \sin \pi t dt$
7.  $\int \operatorname{cosec}^2(5x) dx$
8.  $\int \frac{\tan e^{-x}}{e^x} dx$
9.  $\int \sin^2(4x) dx$
10.  $\int \cos^3(2t) dt$
11.  $\int \frac{x dv}{v^2}$
12.  $\int \frac{e^{2t}}{1+e^{2t}} dt$
13.  $\int \frac{x}{1+2x^2} dx$
14.  $\int \frac{1}{1+2x^2} dx$
15.  $\int \frac{1}{\sqrt{4+8x}} dx$
16.  $\int \frac{x}{\sqrt{1-9x^4}} dx$
17.  $\int \frac{x}{\sqrt{4+2x}} dx$
18.  $\int \frac{x-x^2}{x-1} dx$

### Oplossing / Solution

1.  $\int (1-3x)^2(3+x)^2 dx = \int (9-48x+46x^2+48x^3+9x^4) dx = 9x - \frac{48}{2}x^2 + \frac{46}{3}x^3 + \frac{48}{4}x^4 + \frac{9}{5}x^5 + C$
2.  $\int x^3 \sqrt{1-16x^4} dx = -\frac{1}{64} \times \frac{2}{3} (1-16x^4)^{\frac{3}{2}} + C = -\frac{1}{96} (1-16x^4)^{\frac{3}{2}} + C$
3.  $\int e^{\sin(4t)} \cos(4t) dt = \frac{1}{4} e^{\sin 4t} + C$
4.  $\int \sin(2^{3x}) 2^{3x} dx = -\frac{1}{3 \ln 2} \cos 2^{3x} + C$
5.  $\int \cos \pi dx = (\cos \pi)x + C = -x + C$
6.  $\int \sin \pi t dt = -\frac{1}{\pi} \cos \pi t + C$
7.  $\int \operatorname{cosec}^2(5x) dx = -\frac{1}{5} \cot(5x) + C$
8.  $\int \frac{\tan e^{-x}}{e^x} dx = \int \frac{\sin e^{-x} \times e^{-x}}{\cos e^{-x}} dx = \ln |\cos e^{-x}| + C$
9.  $\int \sin^2(4x) dx = \int \frac{1}{2} [1 - \cos(8x)] dx = \frac{1}{2} [x - \frac{1}{8} \sin(8x)] + C$

$$10. \int \cos^3(2t) dt = \int [1 - \sin^2(2t)] \cos(2t) dt = \int \cos(2t) - \sin^2(2t) \cos(2t) dt = \frac{1}{2} \sin(2t) - \frac{1}{2} \times \frac{1}{3} \sin^3(2t) + C$$

$$11. \int \frac{x dv}{v^2} = x \int v^{-2} dv = x(-v^{-1}) + C = -\frac{x}{v} + C$$

$$12. \int \frac{e^{2t}}{1 + e^{2t}} dt = \frac{1}{2} \ln(1 + e^{2t}) + C$$

$$13. \int \frac{x}{1 + 2x^2} dx = \frac{1}{4} \ln(1 + 2x^2) + C$$

$$14. \int \frac{1}{1 + 2x^2} dx = \frac{1}{\sqrt{2}} \arctan(\sqrt{2}x) + C$$

$$15. \int \frac{1}{\sqrt{4 + 8x}} dx = \frac{2}{8} \sqrt{4 + 8x} + C$$

$$16. \int \frac{x}{\sqrt{1 - 9x^4}} dx = \int \frac{x}{\sqrt{1 - [3x^2]^2}} dx = \frac{1}{6} \arcsin 3x^2 + C$$

$$17. \int \frac{x}{\sqrt{4 + 2x}} dx = \frac{1}{6} (4 + 2x)^{\frac{3}{2}} - 2\sqrt{4 + 2x} + C$$

$$\text{Let } \sqrt{4 + 2x} = t \Rightarrow x = \frac{1}{2}(t^2 - 4) \Rightarrow \frac{dx}{dt} = t$$

$$\int \frac{x}{\sqrt{4 + 2x}} dx = \int \frac{\frac{1}{2}(t^2 - 4)}{t} t dt = \int (\frac{1}{2}t^2 - 2) dt = \frac{1}{2} \times \frac{1}{3} t^3 - 2t + C$$

$$18. \int \frac{x - x^2}{x - 1} dx = \int \frac{x(1 - x)}{x - 1} dx = \int -x dx = -\frac{1}{2} x^2 + C$$